Sabancı Üniversitesi

CS58012 · AGENTIC AI SYSTEMS: ARCHITECTURE AND DEPLOYMENT

Personal Bureaucracy Digital Twin

A stateful agentic assistant for personal bureaucracy

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Goal

Always know where I am,
what is missing,
and what I should do next.

PROBLEM & CONCEPT

Bureaucratic procedures are a dependency problem

Fragmented

Rules are spread across official portals, institutional websites, PDFs and offices.

Dependent

Identity documents, insurance, certificates, payments or address proofs can block later steps.

Changing

Documents can expire, rules can change, and applications can be rejected after the initial plan.

The Digital Twin

Persistent shared state — not another agent.

1. **Where am I now?**
   current documents + completed steps
2. **What is missing?**
   prerequisites + blockers
3. **What should I do next?**
   personalized next action + deadline

Initial pilot / use case

International students in Türkiye · Residence permit (İkamet) · Insurance · Housing / address · University documents

PROJECT TOPIC & AGENCY

Why this needs an agent — and where it can fail

Traditional checklist / single LLM

Same route for everyone

Cannot adapt to different documents, prerequisites, procedures or user situations.

Weak state continuity

Each interaction risks starting from incomplete context.

No grounded recovery

A wrong date, missing prerequisite or rejected document can silently propagate.

Agentic system

Stateful

Reads the user's current administrative state.

Adaptive

Different states and procedures create different next steps.

Feedback-driven

Observe → plan → validate → replan only if needed.

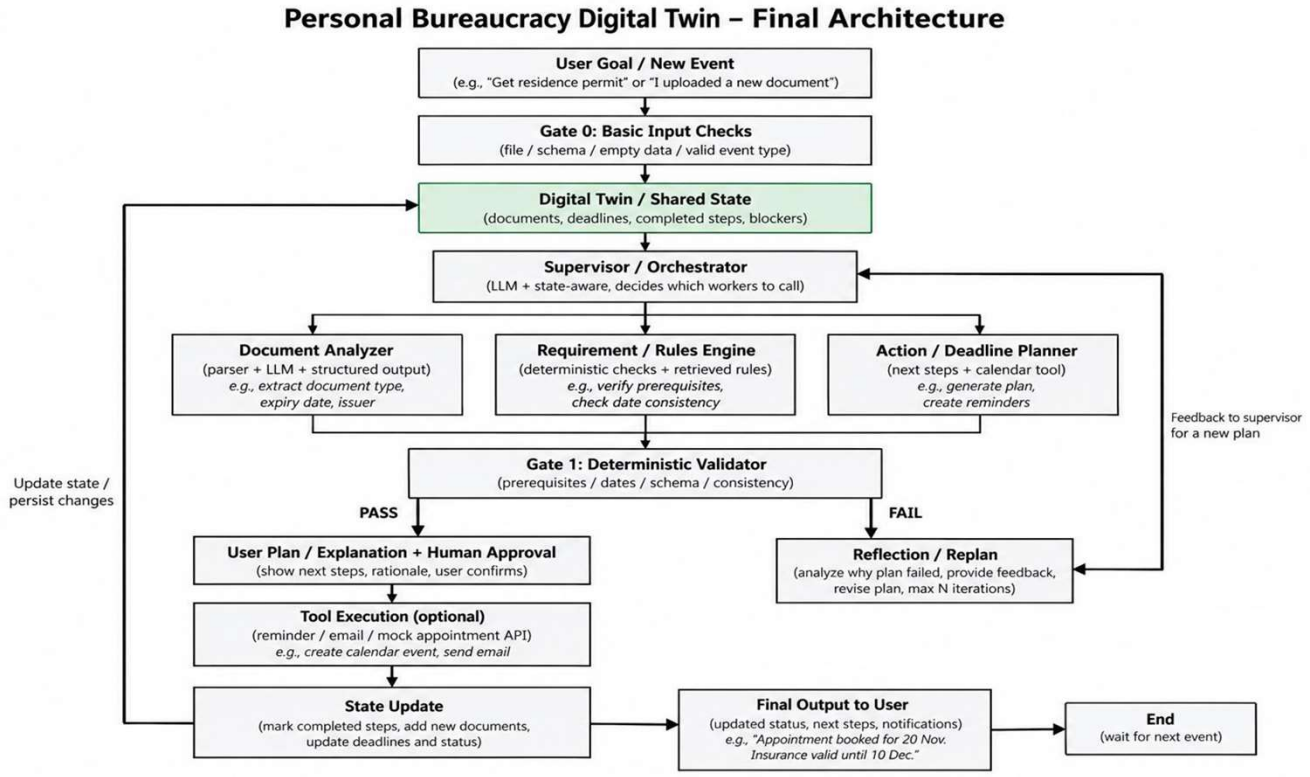
Main failure risks

Hallucinated deadlines • Missing prerequisites • Stale/conflicting rules • State inconsistency • Runaway loops

Design rule: use the minimum autonomy needed; sensitive or consequential actions stay human-approved.

INITIAL AGENT FRAMEWORK

LangGraph + Supervisor / Orchestrator + deterministic gates



Why LangGraph

Cycles, checkpoints and conditional edges make failure → replan explicit.

Why a Supervisor

One bureaucratic goal can require several cross-domain checks; the needed workers are selected at runtime.

PASS / FAIL

PASS → approval, tools, state update
FAIL → grounded reflection, back to Supervisor

Use LLMs for ambiguity; use code for exactness

MAIN PATTERN

Orchestrator–Workers

The Supervisor dynamically decides which specialized checks are needed from the current state.

Deterministic

Gate 0: Basic Input Checks
Gate 1: Deterministic Validator
Requirement Engine
State Update

LLM-assisted

Supervisor / Orchestrator
Document Analyzer
Action Planner
Reflection / Replan

Supporting patterns

- **Prompt chaining** — document extraction
- **Routing** — local branches
- **Parallelization** — independent checks

Shared state documents · dates · active procedures · missing prerequisites · blocked steps · validation errors

Circuit breaker max 2–3 replans → then human review

REFLECTION & MEMORY

Grounded correction now; selective Reflection memory for later episodes

1 **Grounded Critic**

Gate 1 returns a structured error from deterministic evidence.

2 **Bounded Self-Refine**

Repair only the current plan using that error.

3 **Gate 1 re-check**

Accept the correction only after validation.

After a correction passes, the episode can be compressed into one validated lesson.

Digital Twin state

User-specific administrative facts and outcomes

documents · deadlines · completed / rejected steps

Rule / knowledge base

Authoritative requirements

dependencies · source versions · deterministic rules

Reflexion memory

Validated episodic lessons only

failure type · corrective action · outcome

No raw chain-of-thought. If no correction works after the retry limit → stop and ask for human review or missing information.

END-TO-END EXAMPLE

From an uploaded receipt to a corrected plan

1 **Upload receipt**

Receipt_Tuition_2026.pdf
15 Oct · 1500 EUR

2 **Dependency**

Fee unlocks updated
Student Certificate

3 **Planner**

Download certificate
+ keep permit plan

4 **Gate 1 FAIL**

Insurance 10 Nov
Appointment 20 Nov

5 **Replan**

Renew insurance before
20 Nov + reminder

Why the agent matters here

The original plan looked plausible, but a deterministic date check exposed a downstream failure. The system repairs only the affected part instead of restarting the whole procedure.

Final user output Insurance expires before the appointment. Next: renew it before 20 Nov. Then add a reminder and update the Digital Twin.

TAKEAWAYS

Persistent, adaptive assistance — not one-shot Q&A

Stateful

The Digital Twin keeps documents, deadlines, blockers, active procedures and previous outcomes across sessions.

Hybrid

LLMs handle interpretation and planning; deterministic code handles exact rules, dates and validation.

Bounded & verifiable

Reflection starts from grounded evidence, retries are limited, and sensitive actions stay human-approved.

Why Agentic AI?

Because bureaucratic paths depend on persistent state, runtime failures and changing prerequisites — so the system must observe, validate and adapt. We will validate the idea first on the İkamet use case, then extend the rule base to other procedures.

Questions?

Thank you!